M5SHARK

DEVICE OWNER GUIDE

M5SHARK GUIDE

SETUP + SAFETY

UNIVERSAL RF INCLUDED

M5Shark Owner Guide

Setup, device reference, firmware links, safety notes, and troubleshooting for the M5Shark hardware ecosystem.

M5Shark Guide

Complete setup and reference manual

Exact device family

Choose the right firmware path

Lab-first safety

Receive-first and authorization-first

Owner-ready

Links, FAQ, troubleshooting

For lawful education, development, repair, interoperability study, and authorized testing only.

1. How to Use This Guide

This guide helps you identify the right device family, choose the correct firmware path, use accessories safely, and troubleshoot common setup problems. It is written for lawful education, development, repair, interoperability study, and authorized testing.

Start correctly

Identify your exact device family before using firmware, accessories, or setup links.

Use safe workflows

Start with receive-first learning, owned devices, lab demos, and authorized testing only.

Universal RF reference

The Universal RF section is framed for compliance review, shielded-lab study, and safety planning.

Help built in

Includes quick start, firmware paths, FAQ, troubleshooting, and official reference links.

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Device families

Clear product overview for the M5Shark hardware ecosystem.

5

Firmware paths

Flipper-style, Bruce, Marauder, Chameleon, SDR, and board-specific routes.

1

Universal RF guide

Presented as a compliance-review and shielded-lab safety reference.

100 %

Authorization-first

Use only with permission, owned devices, and controlled environments.

2. Product Family Overview

Use this overview to understand what each product family is for, what to prepare before first use, and which safety reminders matter most.

Product family	Best for	Before first use
M5Shark / SHARK RF	Portable RF research and Flipper-style learning device for lawful education.	Confirm exact hardware revision, charge battery, insert microSD, and use compatible firmware only.
ESP32/Bruce/Marauder devices	Open-source ESP32 lab tools for Wi-Fi/BLE/RF education and device learning.	Choose the exact board target before flashing. Use owned networks and lab devices only.
Chameleon Ultra V2	RFID/NFC training, tag technology study, and authorized lab testing.	Use only with tags, cards, and systems you own or are permitted to assess.
H4 + R10 SDR	Receive-first spectrum learning and SDR exploration.	Start receive-first, check antenna selection, and follow local RF rules.
Universal RF Research Tool	Compliance-review, antenna planning, and shielded-lab RF education reference.	Operate only in lawful, approved, controlled, shielded environments with a responsible operator.

What this guide helps you do

For M5Shark device owners:

- Clear device-family guidance so you know exactly which product you have.

- Safe setup paths with firmware links and recovery-first guidance.

- Simple checks before changing firmware: exact model, board revision, About/Firmware screen, cable, and SD card.

- A responsible-use rule that protects you and others: own it, have permission, stay in the lab, stop if unsure.

Responsible-use note

Do not use any device for bypassing systems, attacking public networks, disrupting signals, or using third-party cards/remotes. Keep all activity focused on education, repair, diagnostics, interoperability, and authorized testing.

3. From Box to First Safe Use

Follow this simple path before updating firmware, connecting accessories, or starting any RF, RFID, NFC, IR, Wi-Fi, BLE, SDR, or GPIO workflow.

Step	Owner action	What the guide provides
1. Identify	Check exact product family, board revision, and included accessories.	Device chooser, product details, model and firmware warnings.
2. Inspect	Check battery area, USB-C cable, SD card, antenna connector, enclosure, and screen.	Quick start, safety checklist, heat and charging notes.
3. Setup	Use only the firmware path that matches the exact device family.	Flipper, Bruce, Marauder, Chameleon, SDR, and board-specific links.
4. Learn	Start with receive-first, demo, scan, log, or owned-device workflows.	Mode map, shared menu map, safe lab mode, owner-ready explanations.
5. Troubleshoot	If something fails, check the exact model, firmware screen, cable, SD card, and accessory.	Troubleshooting table, FAQ, and clear owner-friendly explanations.

For beginners

They get simple words: what it is, what it is for, what not to do, and which setup path to use.

For advanced users

They get technical links, firmware choices, and clear compatibility notes.

For troubleshooting

Check the exact model, board revision, cable, SD card, firmware screen, and accessory before changing firmware.

For compliance

The Universal RF section stays framed as authorized shielded-lab and compliance-review reference only.

4. Device Chooser

A clean reference table for the M5Shark hardware ecosystem. It helps you choose the right product family and prevents wrong firmware expectations.

Device family	Type	Best fit / main use
M5Shark Device / SHARK RF Research Device	Portable RF research multi-tool	RF, NFC, RFID, IR, GPIO, authentication demos, hardware learning, and authorized security education.
ESP32-DIV V2.0 Development Board	ESP32-S3 multi-band handheld toolkit	Wi-Fi/BLE learning, 2.4 GHz research, Sub-GHz study, IR lab work, SD logging, and touchscreen demos.
RF CLOWN V2.0	ESP32/nRF24 RF research board	2.4 GHz protocol learning, portable RF experiments, firmware study, OLED menu demonstrations, and classroom RF concepts.
nRFBox-v3	ESP32 wireless RF research tool	2.4 GHz observation, BLE discovery, channel scanning, protocol study, and open-source firmware learning.
Mini ESP32 Marauder	ESP32 Marauder family device	Wi-Fi/BLE education, wireless diagnostics, CTF labs, and portable learning workflows.
ESP32 Marauder	ESP32 Marauder family device	Wireless security training, lab scanning, educational demos, and firmware exploration.
ESP32-C5 Marauder / Flipper expansion module	Expansion board / standalone wireless module	Flipper expansion, ESP wireless learning, GPS-assisted demos, large-screen field viewing, and authorized 433 MHz lab tests.
Chameleon Ultra V2	RFID/NFC reader, writer, emulator family	Authorized RFID/NFC training, card technology study, lab tag testing, and access-control education.
T-Embed CC1101 Plus / TDeck Plus	Advanced ESP32-S3 RF handheld family	Larger-screen RF learning, keyboard/control-heavy workflows, Sub-GHz study, and portable protocol labs.
NM-RF-Hat for Bruce CYD	RF expansion hat	Adds CC1101, nRF24, and PN532-style learning modules to compatible CYD/Bruce setups.
H4 + R10 SDR Device	SDR and portable spectrum platform	Software-defined radio learning, receive-first spectrum viewing, signal education, and firmware exploration.
Universal RF Research Tool / Universal Signal RF platforms	Multi-band lab platforms	Bench learning, controlled shielded-lab review, antenna planning, RF safety education, and product comparison.

5. Device Family Details

Every product family is written with the same owner structure: purpose, hardware notes, safe use, and interface recommendation.

M5Shark Device / SHARK RF Research Device

Owner product note

- **Purpose:** RF, NFC, RFID, IR, GPIO, authentication demos, hardware learning, and authorized security education.

- **Hardware notes:** STM32 64 MHz, 315/433 MHz RF, SMA female antenna port, 1.37 inch monochrome LCD, USB-C, microSD up to 256 GB, 2000 mAh Li-Po, approx. 94 x 44 x 22 mm.

- **Firmware truth:** This device is positioned as a Flipper Zero alternative. Use only firmware confirmed for the exact M5Shark / SHARK hardware revision. Do not install random forks or firmware for a different board.

- **Screen map:** Simple home screen: RF, IR, NFC, RFID, GPIO, Files, plus About/Help and safety prompts.

ESP32-DIV V2.0 Development Board

Owner product note

- **Purpose:** Wi-Fi/BLE learning, 2.4 GHz research, Sub-GHz study, IR lab work, SD logging, and touchscreen demos.

- **Hardware notes:** ESP32-S3, 16 MB flash, 2.8 inch ILI9341 touchscreen, SD slot, USB-C, three nRF24 modules, one CC1101 module, RGB LEDs, buzzer, battery power system.

- **Safe use:** Keep any transmit-capable feature lab-only; verify battery installation, antenna connection, and board revision before flashing.

- **UI idea:** Touchscreen tabs: Wireless, RF, IR, Logs, Settings, Help. Default to receive-first demonstrations.

RF CLOWN V2.0

Owner product note

- **Purpose:** 2.4 GHz protocol learning, portable RF experiments, firmware study, OLED menu demonstrations, and classroom RF concepts.

- **Hardware notes:** ESP32-WROOM-32U, triple nRF24 layout, OLED display, three tactile buttons, NeoPixel status indicator, IPEX antenna connectors.

- **Safe use:** Use receive-first workflows; do not frame the product as a disruption or interference tool.

- **UI idea:** Three-button UI: Back, Select, Next, with a persistent status LED meaning table.

5. Device Family Details - continued

nRFBox-v3

Owner product note

- **Purpose:** 2.4 GHz observation, BLE discovery, channel scanning, protocol study, and open-source firmware learning.
- **Hardware notes:** ESP32 Wroom32U platform, nRF24 wireless modules, OLED interface, onboard controls, compact enclosure.
- **Safe use:** Limit use to owned devices and lab environments; avoid nuisance traffic or disruption.
- **UI idea:** Default channel view: channel, signal level, mode, log status, battery, and help.

Mini ESP32 Marauder

Owner product note

- **Purpose:** Wi-Fi/BLE education, wireless diagnostics, capture-the-flag labs, and portable learning workflows.
- **Hardware notes:** ESP32-based Marauder firmware device; exact controls and screen depend on variant.
- **Safe use:** Use only on networks and devices you own or are authorized to test.
- **UI idea:** Firmware-first map: Wi-Fi, BLE, Files, Logs, Settings, Update.

ESP32 Marauder

Owner product note

- **Purpose:** Wireless security training, lab scanning, educational demos, and firmware exploration.
- **Hardware notes:** ESP32-based device built for the Marauder firmware ecosystem; hardware details vary by build.
- **Safe use:** Keep owner guidance educational and authorization-first.
- **UI idea:** Use the same menu labels across Marauder devices so owners can move between models.

5. Device Family Details - continued

ESP32-C5 Marauder and Flipper Expansion Module

Owner product note

- **Purpose:** Flipper expansion, ESP wireless learning, GPS-assisted demos, large-screen field viewing, and authorized 433 MHz lab tests.

- **Hardware notes:** 4-in-1 expansion concept: ESP module, 433 MHz RF, GPS, 2.8 inch display, external antennas, rechargeable battery depending on version.

- **Safe use:** Confirm compatibility before connecting to another device; protect GPIO pins and avoid unknown targets.

- **UI idea:** Connection-state screens: Standalone, Connected, GPS Ready, RF Ready, Battery.

Chameleon Ultra V2

Owner product note

- **Purpose:** Authorized RFID/NFC training, card technology study, lab tag testing, and access-control education.

- **Hardware notes:** Dual-frequency RFID/NFC ecosystem covering common LF/HF study workflows with USB-C/Bluetooth control depending on build.

- **Safe use:** Only test tags, cards, and systems you own or are permitted to assess.

- **UI idea:** Clear card-state labels: Read, Saved, Emulate, Write, Locked, Error.

T-Embed CC1101 Plus and TDeck Plus

Owner product note

- **Purpose:** Larger-screen RF learning, keyboard/control-heavy workflows, Sub-GHz study, and portable protocol labs.

- **Hardware notes:** ESP32-S3 family with display and RF options; CC1101 and nRF24 compatibility depend on exact build.

- **Safe use:** Match firmware to the exact board and match antennas to the region and frequency band.

- **UI idea:** Mode dashboard with battery, region profile, antenna reminder, storage status, and firmware version.

5. Device Family Details - continued

NM-RF-Hat for Bruce CYD

Owner product note

- **Purpose:** Adds CC1101, nRF24, and PN532-style learning modules to compatible CYD/Bruce setups.
- **Hardware notes:** Expansion hat concept for RF/NFC modules; exact wiring and compatibility depend on CYD board revision.
- **Safe use:** Verify pinout, voltage, and firmware target before powering the hat.
- **UI idea:** Accessory-detect screen: module present, voltage OK, antenna present, storage ready.

H4 + R10 SDR Device

Owner product note

- **Purpose:** Software-defined radio learning, receive-first spectrum viewing, signal education, and firmware exploration.
- **Hardware notes:** H4 + R10 two-in-one SDR device family with display, controls, USB-C or SDR host features depending on model.
- **Safe use:** Keep guidance receive-first unless you have legal authority and understands RF rules.
- **UI idea:** Spectrum screens with frequency, mode, gain, recording status, and legal reminder.

5. Device Family Details - continued

Universal RF Research Tool / Universal Signal RF Platforms

Owner product note

- **Purpose:** Bench learning, controlled shielded-lab review, antenna planning, RF safety education, and product comparison.

- **Hardware notes:** The supplied manual source describes a handheld aluminum unit with cooling fan, internal battery, channel switches, indicator lights, and charging socket.

- **Corrected identity:** The source-labeled source manual should be treated in this owner guide as the Universal RF Research Tool manual reference, not as a separate product family.

- **Safe use:** Compliance review only. RF shielding or interference-capable equipment requires expert legal and technical review before import or operation.

6. Modules, Accessories, and Shared UI/UX

> Make every device feel like one ecosystem: identify the device, confirm safe conditions, choose a mode, save logs/files, and find help.

Modules and accessories

Accessory	Role	Owner note
CC1101 Wireless Module	Sub-GHz RF module	Use 3.3 V logic, correct antenna, and legal bands only.
NRF24L01+PA+LNA Module	2.4 GHz module	Use stable 3.3 V power and an antenna matched to 2.4 GHz.
2.4 GHz Wi-Fi/Bluetooth Antenna	Antenna accessory	Match SMA/RP-SMA connector type and frequency band.
Micro SD Card Module / Storage Board	Storage accessory	Use for logs, configs, profiles, and safe files.
PCB Protoboard	Prototyping board	Use for low-voltage electronics practice and safe bench wiring.

Owner experience guide

Experience area	Recommended experience
First-run onboarding	Welcome > choose device family > safety agreement > battery/storage check > antenna reminder > safe home screen.
Home screen layout	Top status bar, large current mode card, two primary actions, files/log shortcut, settings/help shortcut.
Mode cards	Each mode shows purpose, required accessory, legal reminder, and a safe start button.
Safe Lab Mode	Default guide mode. It favors receive-first, learning, logs, classroom demos, and owned-device tests.
Error states	Use direct messages: antenna missing, storage not ready, low battery, wrong accessory, wrong firmware.
Help flow	About device > model info > firmware version > accessory status > export device report.

Shared menu map

Menu	What you see
Home	Device status, battery, storage, active accessory, quick actions.
Learn	Short explanations for RF, IR, NFC, RFID, GPIO, SDR, and safety basics.
Tools	Device-specific feature groups with safety labels and accessory reminders.
Files	Saved profiles, logs, captures, notes, and import/export.
Settings	Brightness, language, storage, firmware, region profile, reset.
Help	Safety checklist, troubleshooting, help info, glossary.

7. Firmware and Setup Paths

A firmware image for a similar-looking device can break display, buttons, storage, radio modules, or boot behavior. Match the exact device family first.

Device family	Recommended path
M5Shark Device / SHARK	Use the device firmware and firmware path supplied for the SHARK RF device. For Flipper Zero ecosystem firmware, use only a build confirmed for the exact M5Shark / SHARK hardware revision.
Bruce-compatible products	Use Bruce website, Bruce web flasher, Bruce GitHub, or M5Burner where a matching device build exists. Bruce is for compatible ESP32 devices, not for every STM32/SHARK device.
ESP32-DIV V2.0	Use the ESP32-DIV project resources and hardware notes for the exact board revision.
RF CLOWN V2.0	Use RF-Clown releases and board-specific firmware resources.
nRFBox-v3	Use nRFBox project resources and firmware intended for the exact nRFBox version.
ESP32 Marauder family	Use the ESP32 Marauder getting-started guide and matching releases for the exact board.
Chameleon Ultra V2	Use Chameleon Ultra firmware, app, CLI, and documentation resources.
H4 + R10 SDR Device	Use SDR/Mayhem-style documentation appropriate to the exact device revision.
Universal RF Research Tool	No generic firmware promises. Treat the manual as a hardware/owner safety reference and verify exact model, charger, and legal compliance before any lab evaluation.

Before flashing

Before flashing: confirm the exact model, board revision, and current About/Firmware screen before using any firmware link.

8. M5Shark Device - Flipper-Style Firmware Guide

The M5Shark Device is a Flipper-style RF learning device. Use only firmware confirmed for the exact M5Shark / SHARK hardware revision.

Firmware / tool	Best use	How to install / owner guidance
Official Flipper Zero firmware	Best default choice for stable updates, recovery, and standard firmware update and recovery tools.	Use Flipper Mobile App, qFlipper, or Flipper Lab. Choose Release for normal users.
Official Release / RC / Dev channels	Official firmware channels. Release is stable, RC is testing, Dev is newest but may be unstable.	Use qFlipper advanced controls or Flipper Mobile App update channel selection.
Flipper Lab web updater	Browser tool for updates, apps, and file management on compatible browsers.	Connect by USB, open Flipper Lab, select the device, then follow the on-screen firmware prompts.
qFlipper desktop updater	Desktop updater for Windows, macOS, and Linux; useful for backup, restore, update, and recovery.	Install qFlipper, connect with a data USB cable, insert microSD, choose Release, then update.
Unleashed firmware	Popular custom firmware project for advanced users who understand compatibility and local law.	Use the official project release package or installer instructions only.
Momentum firmware	Modern custom firmware based on official firmware, with features from Unleashed and continuity from Xtreme.	Use the Momentum website or repository install instructions; verify the exact device target before flashing.
RogueMaster firmware	Custom firmware with bundled apps/plugins for experienced users.	Use official RogueMaster releases or web installer instructions; back up files first.
Xtreme firmware legacy	Legacy custom firmware. Keep it as a reference link, but guide most users toward current maintained projects.	Use only when you intentionally need a legacy build and understand rollback/recovery.

Owner install flow

- Confirm the exact M5Shark / SHARK hardware revision and use only firmware marked compatible with that exact target.
- Charge the device, insert a known-good microSD card, and use a data-capable USB cable.
- Back up files, keys, remotes, databases, and saved profiles before changing firmware.
- For normal users, start with the official Release channel through qFlipper, Flipper Mobile App, or Flipper Lab.
- For custom firmware, download only from the official project website or GitHub release page, then follow that project's installer instructions.
- After flashing, verify firmware name/version and test screen, buttons, storage, IR, NFC/RFID, and RF menus.
- If an update fails, stop installing custom builds and recover with official qFlipper before trying another package.

9. Bruce-Style Setup Flow

For compatible products that work with Bruce or Bruce accessories, the flow is: choose the exact target, flash safely, test basic functions, then learn the menus.

Option / topic	Owner guidance
Web flasher	Open the Bruce web flasher in Chrome or Edge, connect a compatible device, select the exact target, and follow the browser prompts.
M5Burner	Install M5Burner, connect the compatible device, choose the correct category, search for Bruce, then download and burn the matching firmware.
Local binary	Advanced users may use device-specific binaries and local flashing tools only when they know the exact target and recovery method.
Recovery	If the device does not boot, stop and restore a known-good build for that exact model before trying another image.
Bruce Firmware	Open-source ESP32 firmware used on compatible devices for security education, RF learning, automation practice, and authorized lab workflows.
Main menus	Wi-Fi tools, BLE tools, IR tools, Sub-GHz/RF tools, files, scripts, settings, updates, logs, and device information where available.
File transfer	Use SD card, USB file manager, or the device web interface when available. Keep backups before firmware updates.
Web interface	Some Bruce builds expose a browser interface such as bruce.local when the device and computer are on the same network.
OTA updates	Where available, update from the device settings after connecting Wi-Fi and backing up important files.
Safe lab use	Start with scanning, learning, logging, owned-device IR profiles, and classroom demos before any advanced feature.

Firmware compatibility note

Bruce is only for Bruce-compatible ESP32 products and accessories. M5Shark / SHARK RF devices use the Flipper-style firmware workflow shown in the M5Shark firmware section. Always choose the exact device model before flashing.

10. Universal RF Research Tool Manual

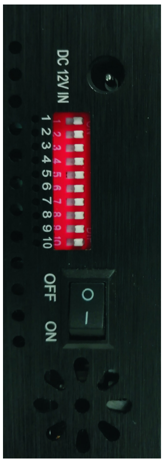
This section uses the correct name Universal RF Research Tool. It is a compliance-review and shielded-lab reference, not an operating guide for public-space interference.

Lab safety note

Use only as a controlled-lab RF education and compliance-review reference. Do not use it to affect public or private communications.

Main unit

Portable unit with antenna bank, aluminum body, and visible power/charge markings.



Control panel

Power switch, charging input, channel indicators, and switch bank.

Cooling side

Side view with aluminum body, cooling area, and venting.

Side/status view

Status area and chassis shape from the supplied manual images.

11. Universal RF Research Tool - Technical Details

Manual information is presented in clear English with legal and safety framing for approved lab review.

Manual topic	Owner guidance
Corrected manual identity	The source manual label is normalized in this guide as Universal RF Research Tool. It belongs inside the Universal RF / Universal Signal RF lab-platform family, not as a separate product listing.
Product description	Portable full-band RF research/compliance-review platform described with surface-mount components and integrated circuits, an aluminum enclosure, active cooling fan, visible power indicator, side controls, antenna/channel switch bank, and charging socket.
Manual-listed evaluation areas	CDMA/GSM/DCS/PHS, 2G/3G/4G/5G, Wi-Fi 2.4 GHz and 5.8 GHz, GPS/BD-GPS, VHF, and UHF. These are manual-listed references only; they do not create permission to transmit, block, or interfere.
Range note	The source manual claims an effective radius around 1-15 m, with some conditions stated as 5-20 m. Real results depend on the local signal environment, antenna set, enclosure, power state, and legal authorization. This guide is not an operating guide for interference.
Power and runtime	Built-in battery, about 2-3 hours runtime according to the source manual, and about 4 hours for a full charge. The manual references AC/home charging and vehicle charging when the supplied adapter allows it.
Physical details	Host size: 165 x 97 x 45 mm. Package size: 240 x 250 x 69 mm. Total kit weight: about 1.11 kg.
Controls and indicators	Cooling fan, main power switch, independent antenna/channel switches, charging socket, and channel indicator lights.
Heat note	The aluminum shell is designed for heat transfer, so normal warmth can occur. Stop using the unit if heat becomes unusual, the fan stops, charging behaves abnormally, or the enclosure is damaged.

12. Universal RF Research Tool - Band References and Safety Screens

> Manual-listed bands are identification references only. They do not create permission to transmit, block, interfere, test public signals, or operate outside a controlled authorized lab.

Manual-listed band groups

Band group	Manual-listed reference
4G / cellular	700 MHz, 2300 MHz, 2600 MHz listed in the manual.
CDMA / GSM / DCS / PHS	800 MHz, 900 MHz, 1800 MHz, 1900 MHz listed in the manual.
3G	2000 MHz and 2100 MHz listed in the manual.
Wi-Fi / 2.4 GHz	2400 MHz listed in the manual.
5G	2600 MHz, 3400 MHz, and 4800 MHz listed in the manual.
Navigation BD/GPS	1200 MHz and GPS 1500 MHz listed in the manual.
VHF / UHF	135 MHz and 470 MHz listed in the manual.
5.8 GHz	5800 MHz listed in the manual.

Safe Universal RF flow

- Confirm the product is lawful to possess, import, and operate in your location before use.
- Use only in an authorized shielded lab, compliance bench, classroom, or controlled test environment where RF interference cannot affect public or private communications.
- Do not operate near emergency services, cellular networks, GPS/GNSS receivers, aviation systems, public safety systems, hospitals, vehicles, schools, or shared public spaces.
- Inspect the enclosure, charger, fan, switches, and accessories before any authorized lab evaluation.
- Keep power off while checking physical accessories and stop immediately if the unit becomes unusually hot or unstable.
- Keep approval records, test location, time, operator, and purpose in the test log.

Screen and safety additions

Screen area	Recommended experience
Compliance gate	Before any channel screen, show: location approved, shielded environment confirmed, responsible operator named, emergency stop available.
Channel status	Use clear labels for each manual-listed band group, but keep all channels off by default until authorized lab conditions are confirmed.
Timer and log	Show a short session timer, approval note, and automatic log entry for start, stop, temperature concern, and power-down events.
Thermal and charging	Expose fan status, battery/charging status, and heat warning messages in the same place on every Universal RF screen.
Stop state	Make the safest state obvious: all channels off, power down, disconnect charger if heat or battery behavior is abnormal, then contact M5Shark help.

13. Device and Accessory Links

Clickable device, firmware, documentation, and technical reference links for setup and learning.

Product / accessory	Setup link
M5Shark Device	Open link https:/ / m5shark.com/ products/ m5shark- device- flipper- zero- alternative
ESP32-DIV V2.0	Open link https:/ / m5shark.com/ products/ esp32- div- v2- 0
RF CLOWN V2.0	Open link https:/ / m5shark.com/ products/ finished- rf- clown- v2- 0- esp32- wireless- security- tester- kit- 3- nrf24l01- ble- bluetooth- analyzer- pentesting- tool- with- oled
nRFBox-v3	Open link https:/ / m5shark.com/ products/ nrfbox- v3- based- on- esp32
Chameleon Ultra V2	Open link https:/ / m5shark.com/ products/ chameleon- ultra- contactless- smart- card- emulator- rfid- smart- chip- reader- 3xcuid- uid- keychain- compliant- to- nfc- read- writer
Mini ESP32 Marauder	Open link https:/ / m5shark.com/ products/ esp32- marauder
ESP32-C5 Marauder	Open link https:/ / m5shark.com/ products/ esp32- c5- marauder- working- with- flipper- zero
ESP32 Marauder	Open link https:/ / m5shark.com/ products/ esp32- marauder- mar- x- auder- device- wi- fi- bluetooth- offensive- and- defensive- tools- enable- secure- reliable- wireless
Universal RF Research Tool	Open link https:/ / m5shark.com/ products/ universal- rf- device- compliance- review
Universal Signal RF 18-Port	Open link https:/ / m5shark.com/ products/ universal- signal- rf- 18- antennas
NM-RF-Hat for Bruce CYD	Open link https:/ / m5shark.com/ products/ nm- rf- hat- for- bruce- cyd
H4 + R10 SDR Device	Open link https:/ / m5shark.com/ products/ h4- r10- two- in- one- sdr- device
T-Embed CC1101 Plus	Open link https:/ / m5shark.com/ products/ t- embed
TDeck Plus	Open link https:/ / m5shark.com/ products/ tsharkplus
CC1101 Wireless Module	Open link https:/ / m5shark.com/ products/ cc1101- wireless- module- with- sma- antenna- wireless- transceiver- module- for- 315- 433- 868- 915mhz
NRF24L01+PA+LNA Module	Open link https:/ / m5shark.com/ products/ nrf24l01- pa- lna- 2- 4g- wireless- module- 1100- meter- long- distance- with- antenna
2.4 GHz Wi-Fi/Bluetooth Antenna	Open link https:/ / m5shark.com/ products/ 3dbi- 2- 4ghz- wifi- antenna- sma- male- router- bluetooth- antennas- wireless- module- 2- 4g- antena- external- aerial- rp- sma- j- spring
Micro SD Card Module	Open link https:/ / m5shark.com/ products/ tf- micro- sd- card- module- mini- sd- card- module- memory- module- for- arduino- arm- avr
PCB Board Protoboard 3x7	Open link https:/ / m5shark.com/ products/ 7x9- 6x8- 5x7- 4x6- 3x7- 2x8cm- double- side- prototype- diy- universal- printed- circuit- pcb- board- protoboard- 4- 6- 6- 8- 5- 7- 3- 7

14. Firmware and Technical Reference Links

Use official firmware sources wherever possible. For custom firmware, use official project pages and verify the exact target.

Resource	Reference link
Bruce website	Open link https:/ / bruce.computer/
Bruce web flasher	Open link https:/ / bruce.computer/ flasher
Bruce firmware GitHub	Open link https:/ / github.com/ BruceDevices/ firmware
Flipper Zero official firmware	Open link https:/ / github.com/ flipperdevices/ flipperzero- firmware
Flipper Zero downloads	Open link https:/ / flipperzero.one/ update
Flipper Lab	Open link https:/ / lab.flipper.net/
qFlipper documentation	Open link https:/ / docs.flipper.net/ zero/ qflipper
Flipper firmware update guide	Open link https:/ / docs.flipper.net/ zero/ basics/ firmware- update
Unleashed firmware	Open link https:/ / github.com/ DarkFlippers/ unleashed- firmware
Momentum firmware	Open link https:/ / github.com/ Next- Flip/ Momentum- Firmware
Momentum website	Open link https:/ / momentum- fw.dev/
RogueMaster firmware	Open link https:/ / github.com/ RogueMaster/ flipperzero- firmware- wPlugins
RogueMaster releases	Open link https:/ / github.com/ RogueMaster/ flipperzero- firmware- wPlugins/ releases
Xtreme firmware legacy	Open link https:/ / github.com/ Flipper- XFW/ Xtreme- Firmware
M5Burner	Open link https:/ / docs.m5stack.com/ en/ uiflow/ m5burner/ intro
ESP32 Marauder getting started	Open link https:/ / github.com/ justcallmekoko/ ESP32Marauder/ wiki/ getting- started
ESP32 Marauder GitHub	Open link https:/ / github.com/ justcallmekoko/ ESP32Marauder
ESP32-DIV hardware wiki	Open link https:/ / github.com/ cifertech/ ESP32- DIV/ wiki/ Hardware
nRFBox GitHub	Open link https:/ / github.com/ cifertech/ nRFBox
RF-Clown releases	Open link https:/ / github.com/ cifertech/ RF- Clown/ releases
Chameleon Ultra GitHub	Open link https:/ / github.com/ RfidResearchGroup/ ChameleonUltra
SDR Mayhem firmware wiki	Open link https:/ / github- wiki- see.page/ m/ portapack- mayhem/ mayhem- firmware/ wiki

Resource	Reference link
CC1101 technical reference	Open link https:/ / www.ti.com/ product/ CC1101
nRF24L01+ specification	Open link https:/ / www.sparkfun.com/ datasheets/ Wireless/ Nordic/ nRF24L01P_ Product_ Specification_ 1_ 0.pdf

Firmware link reminder

Use this reminder under the reference list when sharing the PDF with owners:

- Always pick the firmware target that matches the exact board and hardware revision.
- Release/stable channels are better for normal owners than experimental builds.
- Back up files and saved profiles before updating.
- If a device does not boot after flashing, stop custom attempts and restore a known-good build first.

15. Universal Quick Start and Troubleshooting

A short first-run flow and troubleshooting table for the device owner.

Universal quick start

- Identify the exact device family before using any firmware or accessory.
- Charge the device and inspect cables, antenna, battery area, and enclosure.
- Use only the antenna or accessory intended for the selected mode.
- Start with receive-first or demo modes before any advanced lab workflow.
- Log what was tested, where it was tested, and who approved the test.
- Power down and store the device in its case after use.

Troubleshooting

Problem	First check
Device will not power on	Charge with a known-good 5 V USB source and data/charge cable; inspect the battery area before retrying.
Computer does not detect the device	Use a data-capable cable, another USB port, and the correct driver/firmware tool for the model.
Storage not detected	Reinsert the card, test a known-good card, and confirm the file system is compatible.
Weak or unstable radio behavior	Check antenna type, connector tightness, distance, obstacles, battery state, and local interference.
Wrong menus or missing feature	Confirm the exact firmware build and accessory compatibility for that device family.
Device becomes hot	Stop testing, power down, disconnect charging, and inspect before reuse.
Universal RF tool heat / fan issue	Power down, keep channels off, disconnect charger, let the unit cool, and contact M5Shark help if fan, battery, or enclosure behavior is abnormal.

16. FAQ and Help Checklist

Quick answers for common owner questions, setup problems, firmware confusion, and safe-use reminders.

Question	Recommended answer
Does this work with Flipper Zero firmware?	For the M5Shark / SHARK RF device, only use firmware confirmed for the exact hardware revision. Do not promise that every community fork or older build will work.
Is Bruce firmware for all devices?	No. Bruce is for compatible ESP32 devices and matching board builds. Always select the exact device target before flashing.
Can I test public networks/signals?	No. Use only owned devices, your own lab equipment, or targets where you have clear written permission.
Why are menus different after update?	The wrong build or incompatible version may be installed. Restore the known-good firmware for that exact model, then retest basic functions.
Can the Universal RF Research Tool be used anywhere?	No. It is a compliance-review/lab reference device. Use only where purchase, import, possession, and operation are legal and shielded/authorized.

Before changing firmware

- Confirm the exact device model and hardware revision.
- Use a known-good data USB cable and a charged battery.
- Insert a known-good microSD card when the firmware or apps need storage.
- Check the About/Firmware screen before installing a different build.
- For RF/RFID/NFC issues, verify the accessory, tag, remote, or network is owned by you or authorized for testing.

17. Quick Reference Notes

Short reference blocks for after-purchase setup, safe first use, firmware selection, and Universal RF guidance. Always match the exact product model.

Reference note	Owner guide wording
Short intro	M5Shark devices are designed for lawful cybersecurity education, RF learning, hardware development, diagnostics, repair, and authorized lab testing. Choose your exact product family and use the matching firmware path only.
Firmware note	Firmware compatibility depends on the exact hardware revision. Before updating, check the About/Firmware screen and use only the official or project-compatible target for your device.
Safe use note	Use only with devices, networks, tags, remotes, and signals you own or have permission to test. Do not use in public spaces, shared systems, emergency environments, or any location where testing is not allowed.
Need help note	Need help? Prepare your exact device model, current firmware/About screen, cable/SD card details, and a short description of the issue.
Universal RF note	The Universal RF Research Tool section is provided for authorized shielded-lab RF education and compliance review only. Local laws and permissions must be checked before purchase or use.

18. Flipper-Style Tools and GitHub Resource Directory

Use this section as a safe owner reference for M5Shark / SHARK RF devices that use Flipper-style firmware workflows. These links are for lawful education, firmware management, app discovery, development, repair, and authorized lab testing only.

Safe download rule

Important: do not install random payload packs or unknown files from the internet. Start with official firmware, official apps, and well-known project pages. Community tools can change quickly, so verify the exact hardware target before flashing or copying files.

Resource	Best use for M5Shark owners	Link
Official Flipper Zero firmware	Best default reference for stable firmware, source code, releases, update compatibility, and recovery.	Open GitHub github.com/flipperdevices/flipperzero-firmware
Firmware update guide	Explains release, RC, and dev channels. For normal owners, Release is the safest recommendation.	Open docs docs.flipper.net/zero/basics/firmware-update
qFlipper desktop tool	Use for firmware updates, file management, backup/restore, device information, and recovery on Windows, macOS, and Linux.	Open docs docs.flipper.net/zero/qflipper
qFlipper GitHub	Open-source desktop updater source and issue reference for technical users.	Open GitHub github.com/flipperdevices/qFlipper
Flipper Lab	Browser tool for apps, files, CLI, NFC tools, paint, and compatible web workflows.	Open Lab lab.flipper.net
Apps Catalog	Official place to browse, install, update, and manage apps. Prefer this before third-party app bundles.	Open apps lab.flipper.net/apps
App troubleshooting docs	Use when an app says not found, invalid file, missing imports, outdated app, outdated firmware, or hardware target mismatch.	Open docs docs.flipper.net/zero/apps/troubleshoot
Developer documentation	Reference for app development, JavaScript, firmware internals, hardware modules, and external development workflows.	Open docs developer.flipper.net
uFBT build tool	Official micro build tool for building Flipper apps without cloning the full firmware tree.	Open GitHub github.com/flipperdevices/flipperzero-ufbt
Official IR database	Reference database for legitimate IR remote profiles for owned appliances.	Open GitHub github.com/flipperdevices/IRDB
Awesome Flipper Zero	Large community-maintained index of Flipper firmware, apps, modules, guides, accessories, repair, and reference resources. Verify each item before use.	Open GitHub github.com/djsime1/awesome-flipperzero
Awesome-Flipper website	Community knowledgebase for firmware, apps, modules, cases, and articles in one place.	Open website awesome-flipper.com

19. Flipper-Style Custom Firmware and Compatibility Notes

Custom firmware can be useful for advanced learning, but it can also create app mismatch, storage, boot, or feature problems if it is not built for the exact device target. For most owners, start with stable/release firmware first.

Project	Owner-ready guidance	Link
Unleashed firmware	Popular custom firmware fork for advanced users. Use only from the official project page and only when the target is confirmed compatible.	Open GitHub github.com/DarkFlippers/unleashed-firmware
Momentum firmware	Modern custom firmware option. Verify the exact target and install path before flashing.	Open GitHub github.com/Next-Flip/Momentum-Firmware Open website
RogueMaster firmware	Custom firmware with bundled plugins for experienced users. Back up files first and verify compatibility.	Open GitHub github.com/RogueMaster/flipperzero-firmware-wPlugins
Xtreme firmware legacy	Legacy reference only. Guide most users toward current maintained projects and compatible builds.	Open GitHub github.com/Flipper-XFW/Xtreme-Firmware

Compatibility checklist before any firmware or app install

Use this checklist before installing firmware or apps:

- Exact device model and hardware revision confirmed.
- Battery charged and known-good microSD inserted.
- Data-capable USB cable used for flashing or file transfer.
- Backup completed for saved remotes, NFC/RFID files, keys, logs, and app data.
- Firmware/app target matches the device model and firmware branch.
- If apps fail after a custom firmware update, test again on stable official/release firmware before assuming hardware damage.

20. Flipper Firmware Notice: No Factory Keys Found

This is an information page for owners who see the screen message: “No Factory Keys Found. Secure Enclave is damaged. Some apps will not work.” The message can appear when Flipper-style firmware expects original Flipper factory security keys on compatible M5Shark-style hardware. It does not mean the M5Shark device is physically broken.

Owner question	Clear answer
Is my M5Shark damaged?	No. This warning is normally a firmware compatibility notice, not a sign that the hardware is broken. The device can still boot, open menus, read files, and use normal compatible features.
What should I do on the screen?	Press OK / Back / Continue to skip the warning and enter the main menu. Do not panic and do not keep reinstalling random firmware builds.
Will everything work?	The normal M5Shark functions should work when the correct firmware, SD card, cable, and accessories are used. Some official Flipper security apps that require original factory keys may show limits, but this does not stop the main device workflow.
Why does the message appear?	Some Flipper-style firmware checks for original Flipper factory keys or secure-enclave data. A compatible M5Shark device may not use the same factory-key system, so the firmware shows this notice.
Best first check	Open About / Firmware after skipping the warning. Confirm the firmware name, version, device model, microSD status, battery, and basic button/menu operation.
What not to do	Do not install unknown key-replacement tools and do not flash random firmware repeatedly. Use only the correct firmware path for the exact M5Shark / SHARK hardware revision.

Fast owner steps

Use this simple flow when the warning appears:

- Press OK / Back / Continue to enter the main menu.

- Check that buttons, screen, microSD, Files menu, RF/IR/NFC/RFID menus, and basic settings open normally.

- Use the firmware recommended for your exact M5Shark hardware revision.

- If only one app fails, treat it as an app/firmware compatibility issue, not device damage.

- Contact M5Shark help only if the device cannot boot, buttons do not work, storage is not detected, or the same warning blocks normal menus.

What this message means

Simple owner explanation:

- Your M5Shark is not damaged. This message is a Flipper-style firmware notice about factory keys / secure enclave compatibility.

- Please click OK to skip the message and continue to the main menu. The normal M5Shark functions should work fine with the correct firmware and accessories.

- Some apps that require original Flipper factory keys may show this warning, but this does not mean the device is broken.

- If the main menu still does not open after clicking OK, check the USB cable, SD card, firmware version, and battery, then contact M5Shark with a photo of the About/Firmware screen.

21. Safety and Legal Notice

This guide is for lawful education, development, repair, interoperability study, and authorized testing. It is not permission to test systems, signals, cards, remotes, locks, computers, or networks without authorization.

Owner responsibility

You are responsible for local laws, radio-frequency rules, access-control rules, privacy rules, import restrictions, battery safety, and permission to test. If you are not sure whether a test is allowed, do not run it until you have permission and understand the rules.

Simple rule

Own it. Have permission. Use the right antenna/accessory. Stay in the lab. Stop if the device behaves unexpectedly.

Universal RF rule

The Universal RF Research Tool section is a safety reference for controlled, authorized lab use only. It must not be used to affect public/private communications or safety systems.

Final owner checklist

- Identify the exact M5Shark device family before using firmware or accessories.
- Use only owned or authorized devices, tags, remotes, networks, signals, and lab equipment.
- Start with receive-first learning, safe demo modes, logs, and basic checks.
- Back up files before firmware updates and use the correct target for your hardware revision.
- Stop immediately if the device overheats, charging behaves abnormally, or a test is not clearly allowed.